LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Definition of streamline, stream tube, stream filament, ideal and real fluid, steady and turbulent flow, compressible and incompressible flow
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Definition of streamline, stream tube, stream filament, ideal and real fluid, steady and turbulent flow, compressible and incompressible flow, viscous and in viscous flow, rotational and irrotational flow.
5. Specific Objectives	6.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of flow diagrams from IB; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of air in an aircraft's flight envelope?
(b) Can you list three fundamental properties of a fluid?
(c) Why is understanding flow patterns critical for aerodynamic performance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of streamline • Concept of stream tube • Definition of stream filament • Distinction between ideal and real fluid • Characteristics of steady flow	L & Ds	PPT, IB	15 min
(b)	• Difference between compressible and incompressible flow • Definition of viscous and inviscid flow • Concept of rotational and irrotational flow • Characteristics of turbulent flow • Comparison of flow regimes	L & Ds	PPT, PDF	20 min
(c)	• Application of flow definitions to atmosphere • Summary of key flow classifications • Link to boundary layer formation • Review of previous concepts • Preparation for flow over bodies	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a streamline and a stream tube.
(b) Why is flow considered compressible at high speeds?
(c) How does viscosity affect the classification of real fluid flow?

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of streamline and stream tube will be applied when analyzing flow over bluff bodies in the next lesson.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Flow over bodies; bluff bodies and streamlined bodies, Flow pattern over flat plate, formation of eddies, the wake and its reduction.	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Captain, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min